3.A Compounds and the Mole I

Chemical Formulas. We've already noted that chemical formulas enable us to determine the numbers and types of atoms within compounds. However, not all chemical formulas are the same—some compounds contain ions arranged in an infinite lattice, while others contain discrete molecules comprising atoms linked by covalent bonds. Here, we'll learn how to distinguish between molecular (covalent) and ionic compounds using chemical formulas.

- **Chemical formulas** combine chemical symbols with subscripted numbers, parentheses, and other symbols to represent chemical compounds.

- The group of atoms represented by a chemical formula is called a **formula unit**. Depending on the structure of the substance, the formula unit may correspond to a **molecule** or a **repeating unit** in an infinite lattice.

$$Fe_2(CO_3)_3$$

- Substances vary in the nature of bonding between atoms or ions and the corresponding submicroscopic picture.

    - **Molecular (covalent) compounds** consist of discrete molecules, each of which contains atoms linked by covalent bonds. Electrons are shared between atoms in these bonds. Some elements are molecular—see the next page! Molecular compounds typically contain nonmetal atoms.

    - **Ionic compounds** contain oppositely charged ions held together by electrostatic attraction. Ions are arranged into an effectively infinite lattice in ionic solids. Ionic compounds typically contain metal *and* nonmetal atoms.

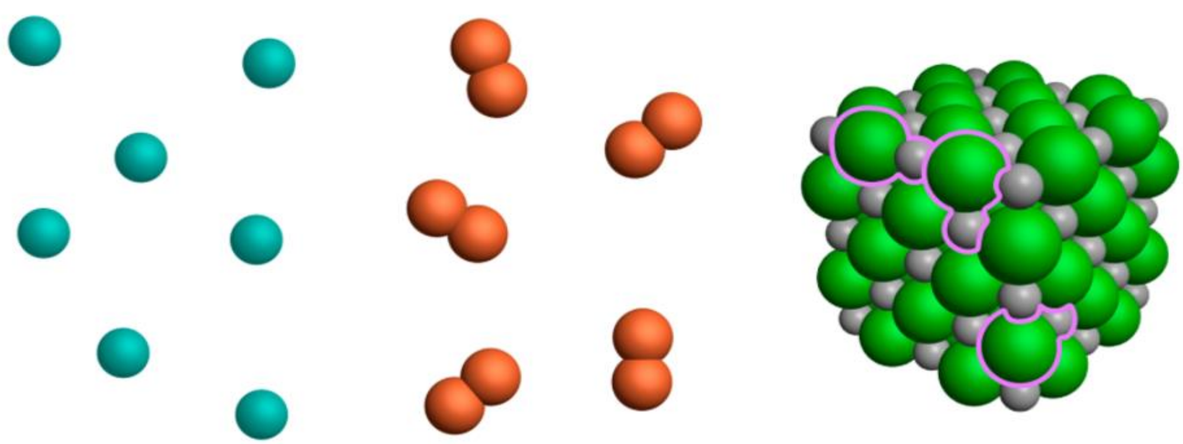


Figure 3.1. Atoms of argon (Ar), molecules of dibromine (Br_2), and an ionic lattice of sodium chloride (NaCl).

- Some elements exist in nature in molecular form, as small molecules containing two or more of the same atom bonded together. Examples include Cl_2, P_4, and S_8.

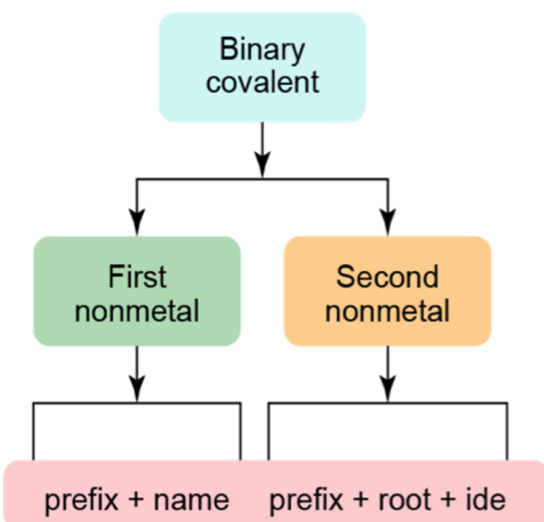
- Seven nonmetallic elements exist in nature as **diatomic** molecules: Br_2, I_2, N_2, Cl_2, H_2, O_2, and F_2. “Brinklehoff!”

Nomenclature of Simple Ionic and Covalent Compounds. Systems of chemical nomenclature enable us to “encode” molecular structure in short and unique names. Although it may seem like the rules of nomenclature are arbitrary, this isn’t quite true—naming conventions are based on patterns in structure. Here, we’ll learn how to name simple ionic and molecular compounds, noting the general “positive negative” convention that pervades names of these substances.

- Molecular (covalent) compounds are composed of two or more *nonmetals*; **binary** covalent compounds contain only two elements. To name binary covalent compounds…

    1. List the first nonmetal with a prefix indicating the number of atoms of that type in the molecule.

    2. After a space, list the second nonmetal with a similar prefix, replacing the ending of the element name with -ide.



Prefix	Number of Atoms per Molecule
mono	1
di	2
tri	3
tetra	4
penta	5
hexa	6
hepta	7
octa	8
nona	9
deca	10

3.1. What is the chemical formula for disulfur hexoxide?

- **Ions** are species with a lack or excess of electrons, which therefore have net positive charge (**cations**) or negative charge (**anions**).

- You should become familiar with the charges of stable monatomic ions of the elements in the main group (1, 2, 13 – 18). *Note the pattern!*

1 1A	2 2A	3	4	5	6	7	8	9	10	11	12	13 3A	14 4A	15 5A	16 6A	17 7A	18 8A
H +1, –1																	
Li +1	Be +2													N –3	O –2	F –1	
Na +1	Mg +2											Al +3		P –3	S –2	Cl –1	
K +1	Ca +2		Ti +2, +3	V +2, +3	Cr +2, +3	Mn +2, +3	Fe +2, +3	Co +2, +3	Ni +2, +3, +4	Cu +1, +2	Zn +2				Se –2	Br –1	
Rb +1	Sr +2								Pd +2, +4	Ag +1	Cd +2		Sn +2, +4			I –1	
Cs +1	Ba +2								Pt +2, +4	Au +1, +3	Hg +2		Pb +2, +4				
Fr +1	Ra +2																

- **Ionic compounds** are composed of metal and nonmetal atoms in ionic form; **binary** ionic compounds consist of one type of cation and one type of anion. Either or both may be polyatomic.

$$Na_2O \qquad\qquad NH_4Cl$$

- Ionic compounds are electrically neutral overall; we can use this idea to deduce chemical formulas or charges.

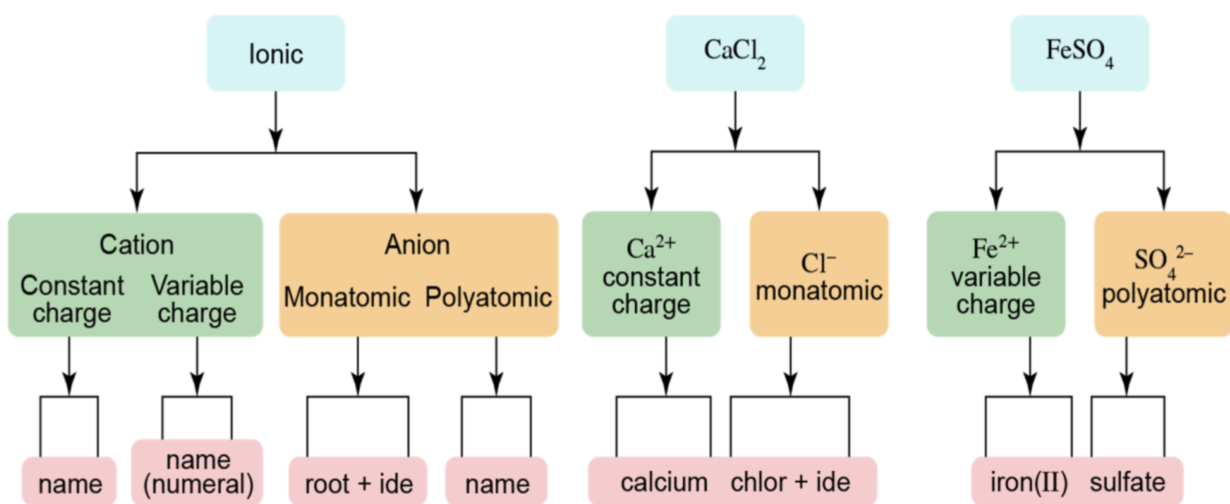
$$Sr_3(PO_4)_2$$

- **Polyatomic ions** contain more than one type of atom; bonds *within* a polyatomic ion are covalent and the net charge may be positive (lack of electrons) or negative (excess of electrons).

Name	Formula	Name	Formula
acetate	$C_2H_3O_2^-$	hypochlorite	ClO^-
ammonium	NH_4^+	chlorite	ClO_2^-
carbonate	CO_3^{2-}	chlorate	ClO_3^-
cyanide	CN^-	perchlorate	ClO_4^-
hydrogen carbonate	HCO_3^-	chromate	CrO_4^{2-}
hydrogen phosphate	HPO_4^{2-}	dichromate	$Cr_2O_7^{2-}$
hydrogen sulfate	HSO_4^-	permanganate	MnO_4^-
hydroxide	OH^-		
nitrite | nitrate	NO_2^- | NO_3^-		
peroxide	O_2^{2-}		
phosphite | phosphate	PO_3^{3-} | PO_4^{3-}		
sulfite | sulfate	SO_3^{2-} | SO_4^{2-}		

- Because ionic compounds are electrically neutral and ion names include information about charge, prefixes are *omitted* in names of binary ionic compounds.

- Ions with variable charge require a Roman numeral in parentheses after the name of the ion.



(a) (b) (c)

3.2. How many oxide ions are required to form an ionic compound with the stable monatomic ion formed by calcium?

3.3. How many sulfite ions are required to form a neutral ionic compound with the ion formed when vanadium has 20 electrons?

Name the chemical compounds below.

$Ca(MnO_4)_2$

SF_6

$Cu(CN)_2$

Ag_2CrO_4

P_4O_{10}

3.B Compounds and the Mole II

Nomenclature Continued. Acids are an important class of covalent compounds that are named somewhat differently from other covalent compounds. Here we'll learn conventions for naming two classes of acids, binary acids and oxyacids.

- **Acids** are a special class of hydrogen-containing covalent compounds that, when dissolved in water…

$$HA(aq) + H_2O(l) \rightleftarrows A^-(aq) + H_3O^+(aq)$$

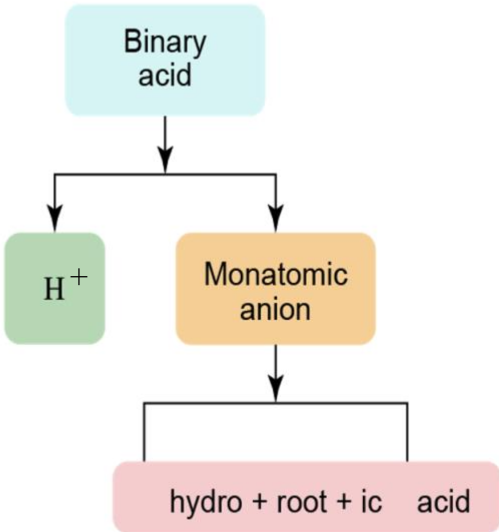
$$HA(aq) \rightleftarrows A^-(aq) + H^+(aq)$$

- **Ionizable hydrogens** are those within an acid that can be lost as H^+.

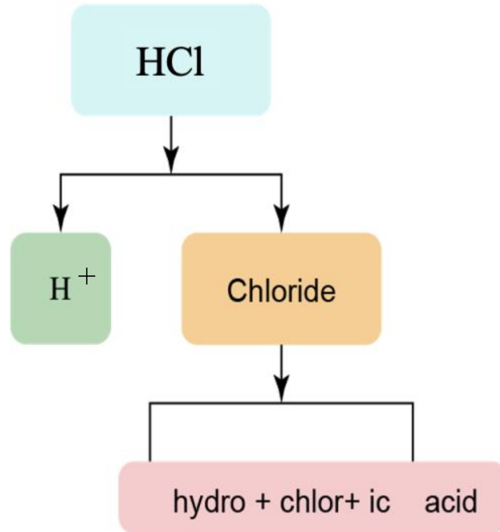
- We will be concerned with two types of acids: **binary acids H_nX** and **oxyacids H_nXO_m**.

- **Examples**
    - HCl is a binary acid.
    - $HClO_4$ is an oxyacid.

- **Binary acids** contain hydrogen and one other element, typically from group 16 or 17.

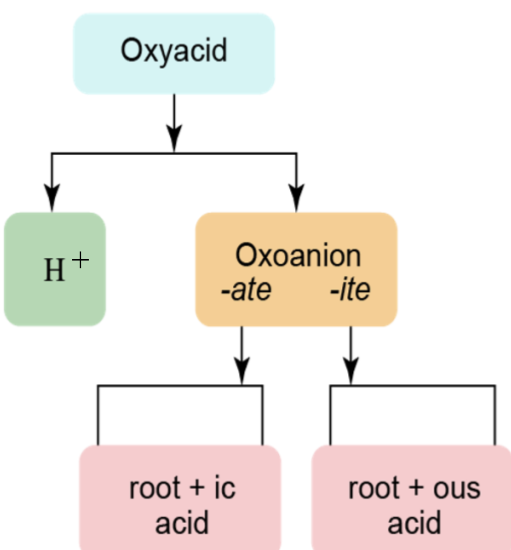


(a)

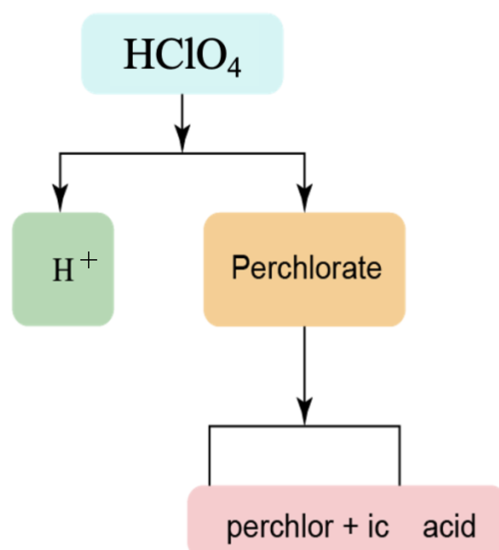


(b)

- **Oxyacids** contain protons (H^+) bonded to polyatomic anions containing oxygen (oxyanions XO_m^{n-}).



(a)



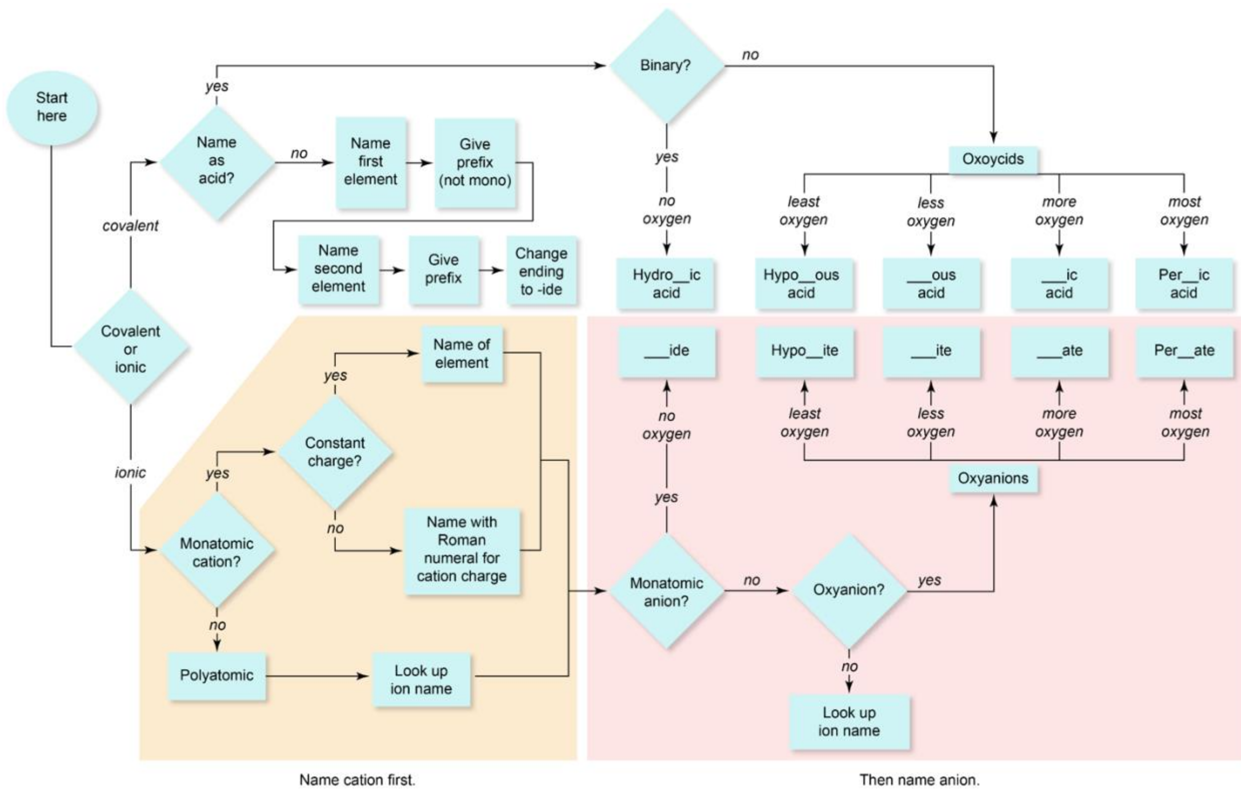
(b)

3.4. What is the name of the acid HNO_2?

A. hydronitric acid
B. hydronitrous acid
C. nitrous acid
D. nitric acid

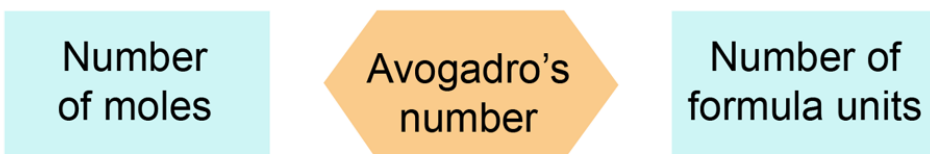
3.5. What is the name of the acid $HBrO_4$?

Review of Nomenclature. The flowchart below includes all the nomenclature conventions we've learned so far. It may be worthwhile to add a version of this to your crib sheet…



The Mole. Chemists face a conundrum when trying to bridge the gap between the mental world of atoms and molecules and the real world of the laboratory. In the mental world, so much of chemistry is based on *number* of molecules, but in the laboratory we measure amount using quantities like mass and volume. We'd like to be able to use mass to count molecules, but to do so, we need to define a special counting unit: the mole.

- The **mole (mol)** is a unit used to count the huge number of molecules or formula units in macroscopic samples.
  - Defined as the number of atoms in exactly 12.0 grams of carbon-12.
  - This definition ensures that 1 mol of a substance with formula mass *M* atomic mass units weighs *M* grams.
  - One mole contains **Avogadro's number (N_A)** of molecules, formula units, or other submicroscopic entity: 6.022×10^{23} mol^{-1}. This is really, really big!



- The **atomic mass** of each element is the average mass of one atom, expressed in atomic mass units (u or amu).
- The formula mass of a compound in atomic mass units (**molecular weight**) is equal to the mass of one mole of the compound (**molar mass**) in grams per mole.
- To find the molar mass of a compound, we simply add up the atomic masses of all atoms in one formula unit of the compound.
- **Example.** Determine the molar mass of C_2H_6O.
  Relevant molar masses: C = 12.01 g/mol; H = 1.01 g/mol; O = 16.00 g/mol.

- Molar mass (g/mol) enables us to…
    - Determine the number of moles corresponding to a given mass of substance
    - Determine the mass of a given number of moles of substance

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3.6. What is the count of oxygen atoms in a sample of 2.67 moles of hydrogen peroxide (H_2O_2)? Enter your response as the base-10 logarithm of this number to the nearest 0.01.

3.7. How many atoms of oxygen are present in 255 g of potassium dichromate?
Determine the chemical formula first!

Enter your response as the base-10 logarithm of this number to the nearest 0.01.

3.8. Calculate the molar mass in g/mol of a covalent compound if the mass of a single molecule is 5.34×10^{-23} g.

Enter your response to the nearest 0.01 g/mol.

3.C Compounds and the Mole III

Molar Mass Continued. When dealing with the masses of submicroscopic collections of atoms (or a mole thereof), we use slightly different terms for ionic and covalent compounds.

- Some notes on vocabulary…
  - **Formula mass (weight)** is a generic term that may refer to an atoms, molecule, or formula unit.
  - **(Average) atomic mass (weight)** refers specifically to atoms.
  - **Molecular mass (weight)** refers specifically to molecules, although this term is sometimes loosely—and incorrectly—applied to formula units of ionic compounds.
  - **Molar mass** refers to the mass of one mole of atoms, molecules, or formula units.

- We'll use the last two terms most commonly, although the distinction between molecular and ionic compounds will remain important throughout CHEM 1211K and 1212K!

From Elemental Masses to Chemical Formulas. It's now clear that, using molar mass, we can use masses to count atoms and molecules. From the masses of the elements within a compound, we can infer the numbers of each type of atom in the compound and the number ratios of those atoms in the compound. In other words, we can work from the masses of the elements in a compound to the chemical formula.

- **Percent composition** refers to the *mass* percentage of each element within a compound.

- If the chemical formula is known, percent composition can be deduced from the formula and the average atomic masses of the constituent elements.

- **Example.** What are the mass percentages of carbon and hydrogen in methane (CH_4)? The molar mass of carbon is 12.011 g/mol and that of hydrogen is 1.008 g/mol.

- Mass percent composition only enables us to determine the formula of a compound as the lowest whole-number ratio of the elements.

    - **Chemical formula** is a general term referring to the numbers and types of the atoms in a compound.

    - **Empirical formula** includes the lowest or simplest whole-number ratios of the elements in the compound.

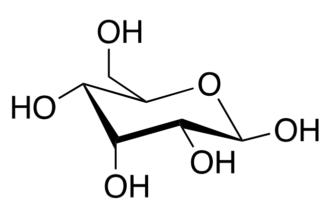
    - **Molecular formula** refers to the composition of a single molecule or formula unit of the substance.

- The empirical and molecular formulas of a compound frequently differ. Consider ethylene, which has the molecular formula C_2H_4.

- To determine an empirical formula from given mass percentages of the elements…
    1. Convert percentage data into mass data, assuming a basis of 100.0 g of substance.
    2. Determine the number of moles corresponding to the mass of each element.
    3. Divide the moles of all elements by the smallest number of moles.
    4. If the resulting numbers are not all whole numbers, scale up by an appropriate factor to make all whole numbers.
    5. Insert the resulting numbers as subscripts into the chemical formula.

- The **molecular formula** contains the composition of one molecule of a molecular compound. Molecular formula is necessarily a multiple of empirical formula (the multiplier may be 1).

- For example, one molecule of glucose contains 6 C atoms, 12 H atoms, and 6 O atoms. The molecular formula is $C_6H_{12}O_6$. What is the empirical formula?



glucose

- **Example.** Vitamin C contains the elements C, H, and O. It is known to contain 40.9% C, 54.5% O, and 4.58% H by mass. The mass of 0.500 moles of vitamin C is 88.07 g. How many atoms of carbon are in a molecule of vitamin C?
    - First, we'll use the given mass percentages to determine the empirical formula of vitamin C.
    - Then, we'll deduce the molar mass, formula mass, and *molecular* formula.

3.9. Determine the empirical formula of acetic acid, which consists of 40.00% carbon, 6.71% hydrogen, and 53.29% oxygen by mass.

3.10. A mass of 6.01 grams of acetic acid corresponds to 0.100 mol of this substance. What is the molecular formula of acetic acid?

Applying Number Ratios in Chemical Formulas. We've already seen that subscripts in chemical formulas help us count atoms in one formula unit of a compound. They can also be used to count *moles* of an element in *one mole* of the compound.

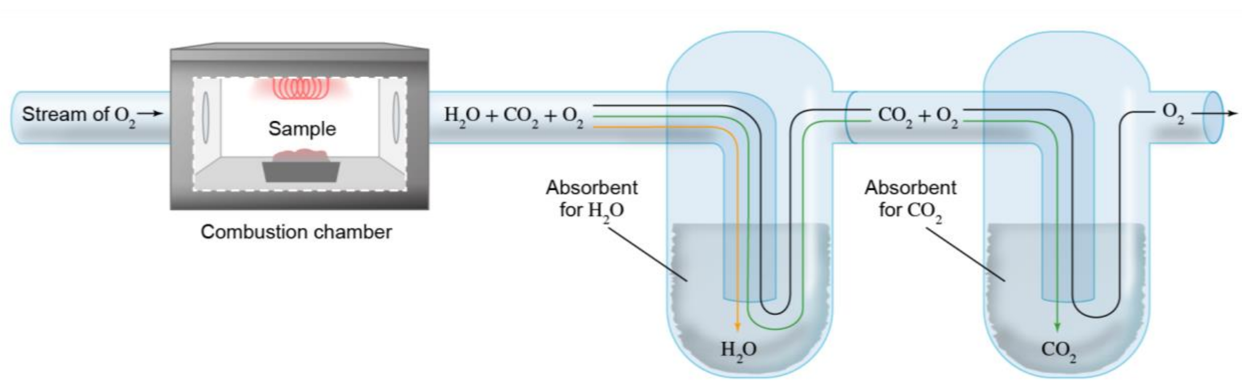
- For example, consider theobromine ($C_7H_8N_4O_2$), an active ingredient in chocolate. If you eat 1 mol of theobromine, you have consumed 7 mol of carbon atoms.

3.11. What is the mass of chromium (chromium *atoms*) in 225 g of potassium dichromate?
Determine the chemical formula first!

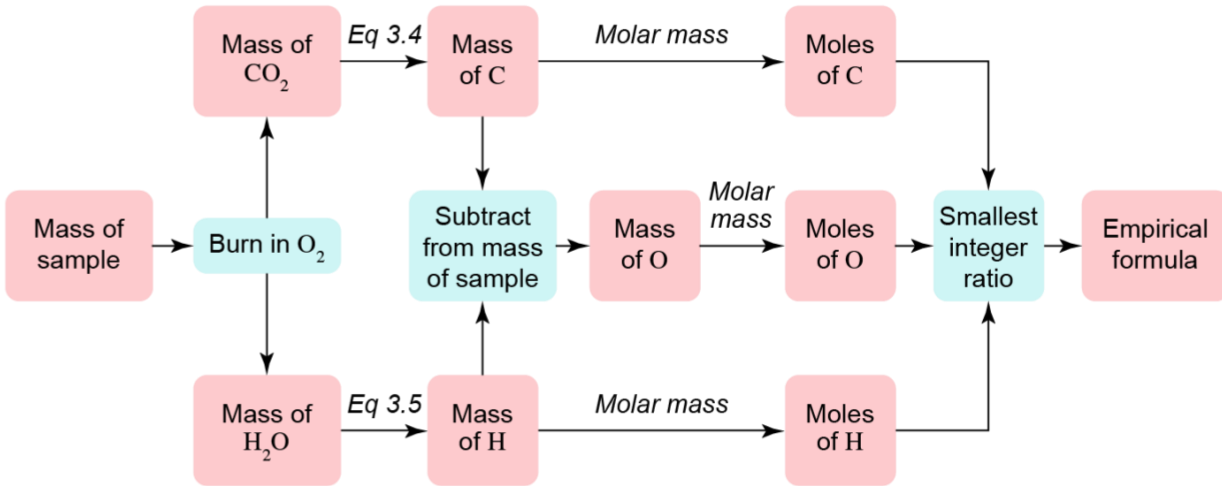
Combustion Analysis. There are a number of practical concerns involved in separating a compound into its component elements. One strategy for compounds containing carbon, hydrogen, and possibly one other element involves burning the compound in oxygen and measuring the masses of CO_2, H_2O, and other oxides produced. From these masses, we can infer the empirical formula of the compound.

- **Combustion analysis** is a laboratory technique used to determine empirical formula of compounds containing carbon, hydrogen, and perhaps a third element.

- A known mass of sample reacts with molecular oxygen (O_2).



- To solve combustion analysis problems, we apply the ideas above using masses of CO_2 and H_2O instead of the elements themselves.



3.12. A 0.250 g sample of hydrocarbon (C_xH_y) undergoes complete combustion to produce 0.845 g of CO_2 and 0.173 g of H_2O. What is the empirical formula of this compound?

Chapter Summary. Some of the amazing things we can now do with compounds and the mole…

1. Distinguish between molecular and ionic compounds using chemical formulas.
2. Visualize molecular and ionic compounds on the submicroscopic level.
3. Name binary covalent compounds; translate the name of a binary covalent compound into a chemical formula.
4. Name binary ionic compounds; translate the name of a binary ionic compound into a chemical formula.
5. Name binary acids; translate the name of a binary acid into a chemical formula.
6. Name oxyacids; translate the name of an oxyacid into a chemical formula.
7. Determine the molar mass of an element or compound from its chemical formula.
8. Apply molar mass to “convert” between mass and moles.
9. Infer the empirical formula of a compound from the mass percentages of the elements within that compound.
10. Apply chemical formulas to infer numbers or masses of atoms within a compound.
11. Use the results of a combustion analysis experiment to infer the empirical formula of a compound containing carbon, hydrogen, and perhaps one other element.